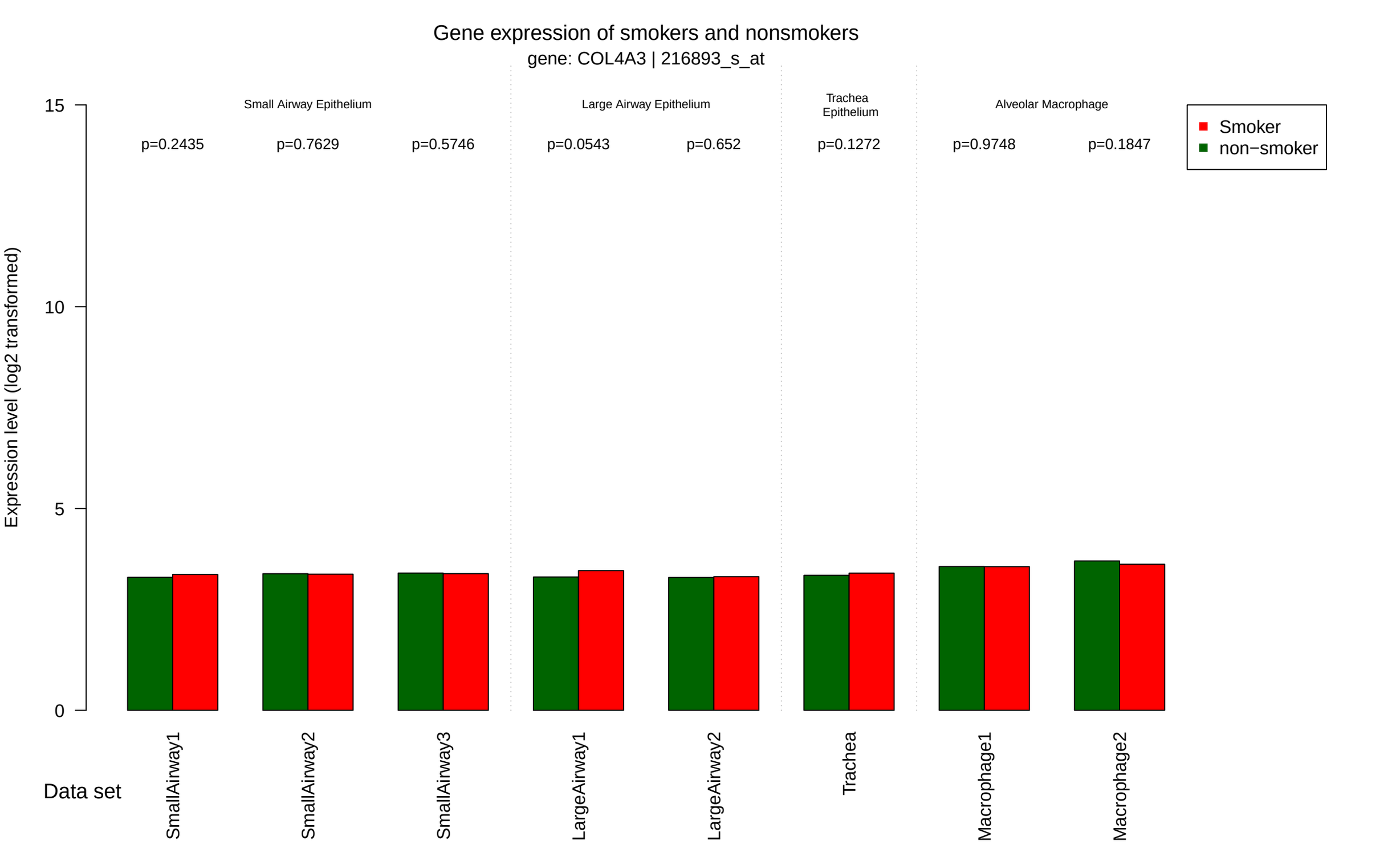
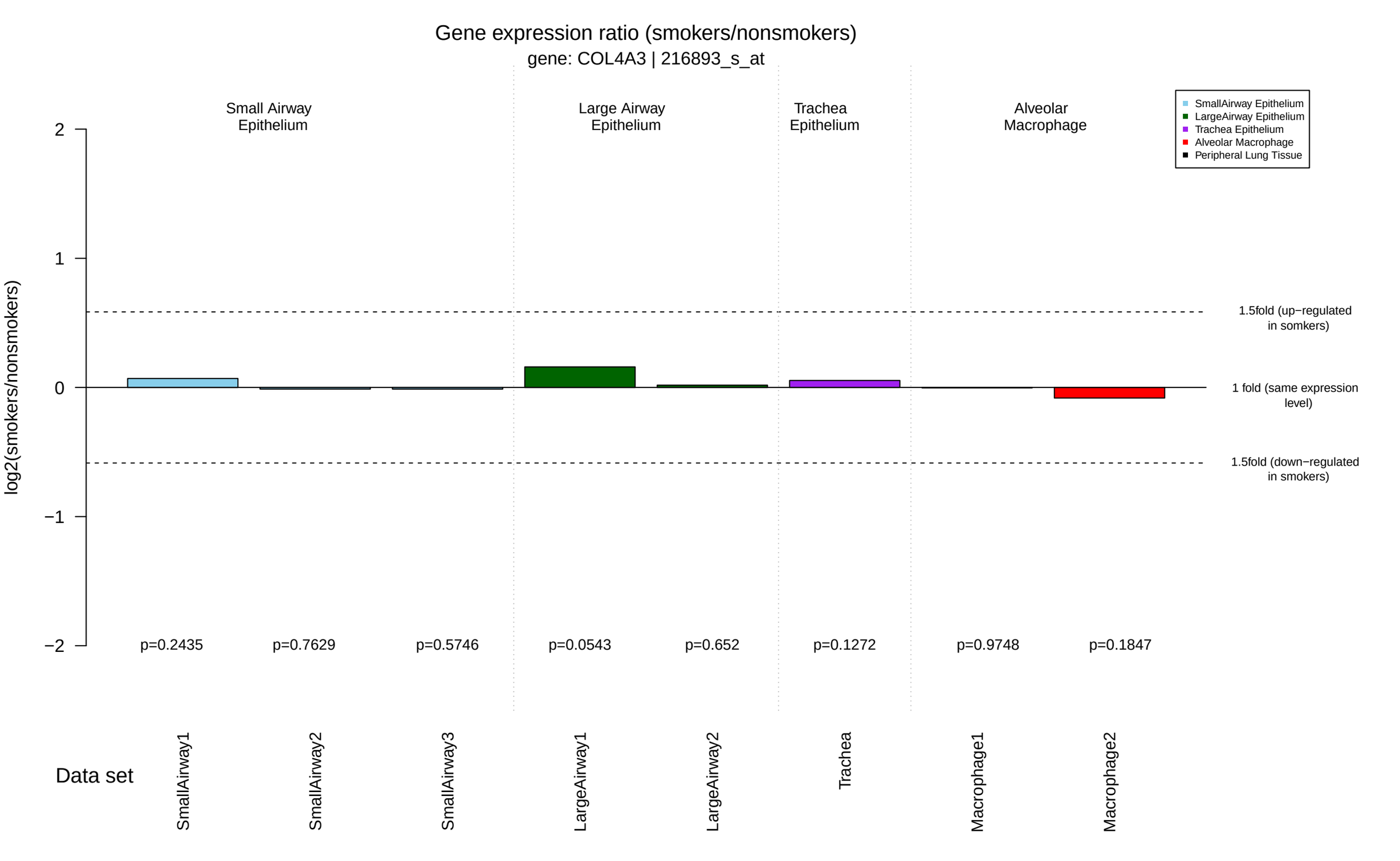
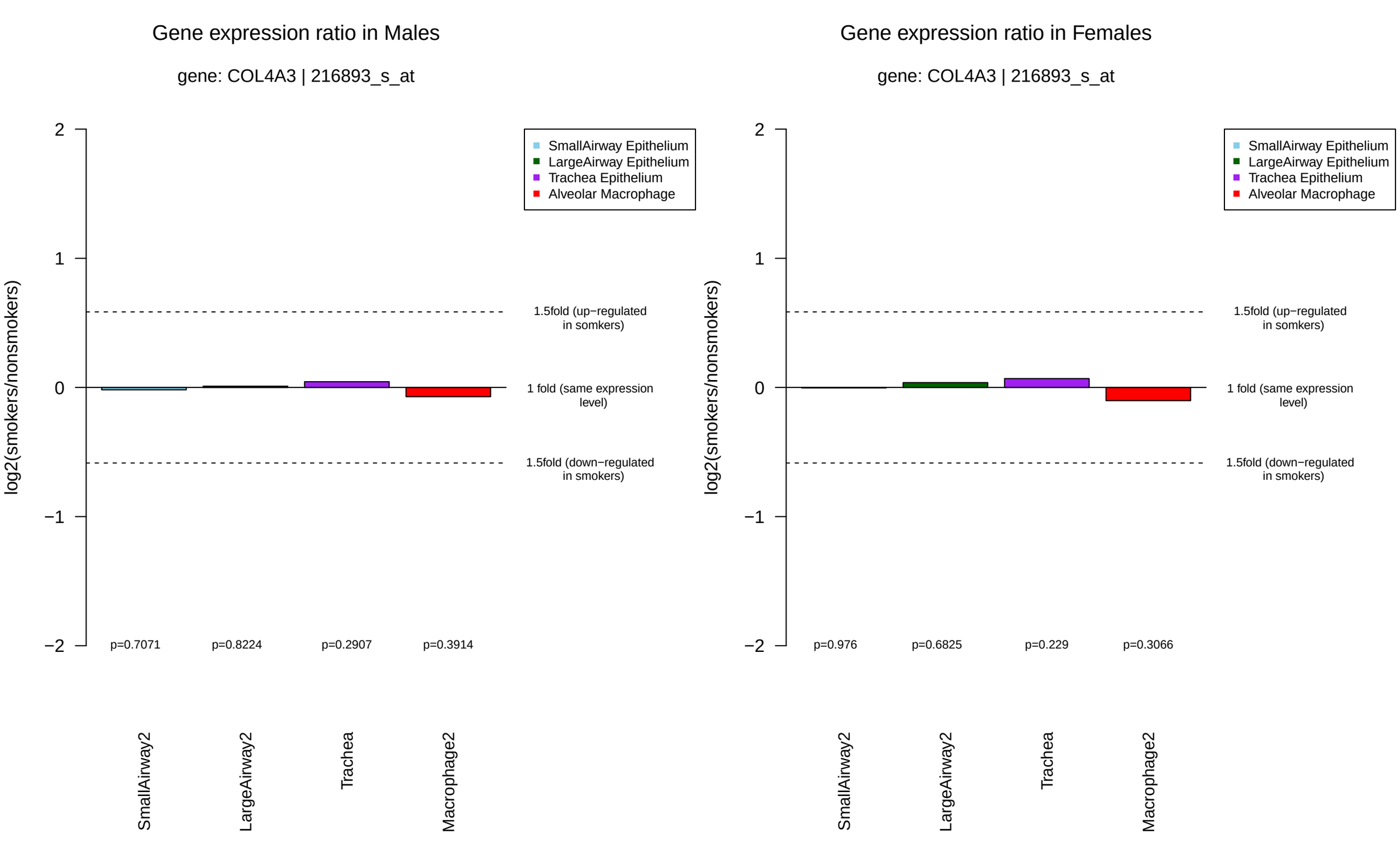
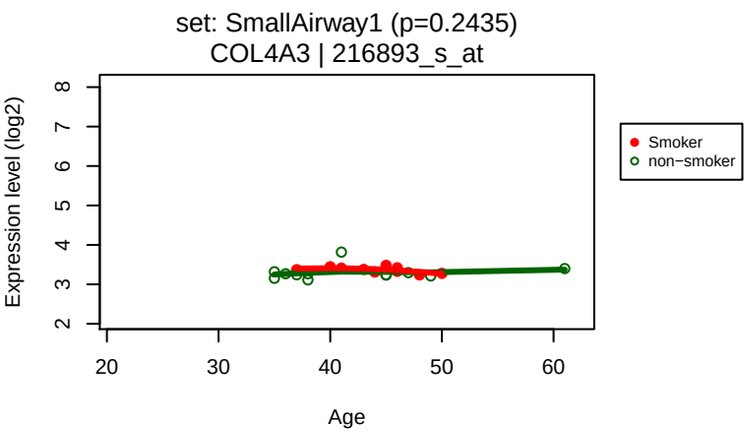
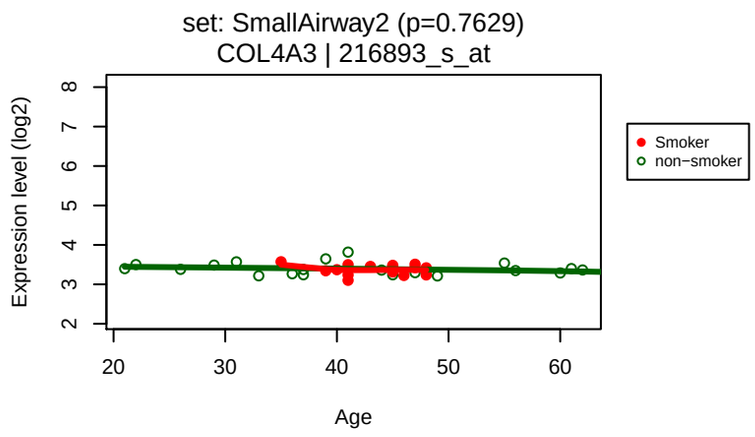
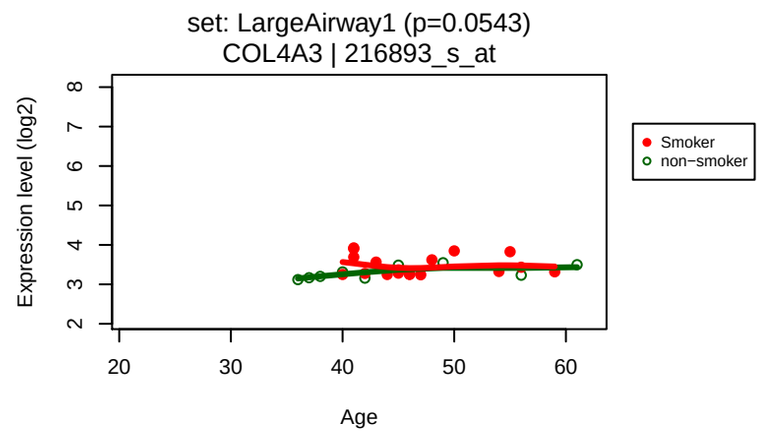

Gene expression of smokers and nonsmokers
gene: COL4A3 | 216893_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2435
p=0.7629
p=0.5746
p=0.0543
p=0.652
p=0.1272
p=0.9748
p=0.1847
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

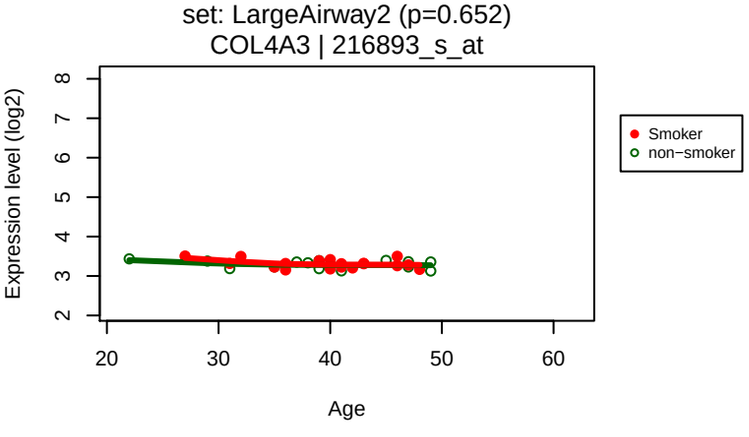

Gene expression ratio (smokers/nonsmokers)
gene: COL4A3 | 216893_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2435
p=0.7629
p=0.5746
p=0.0543
p=0.652
p=0.1272
p=0.9748
p=0.1847
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

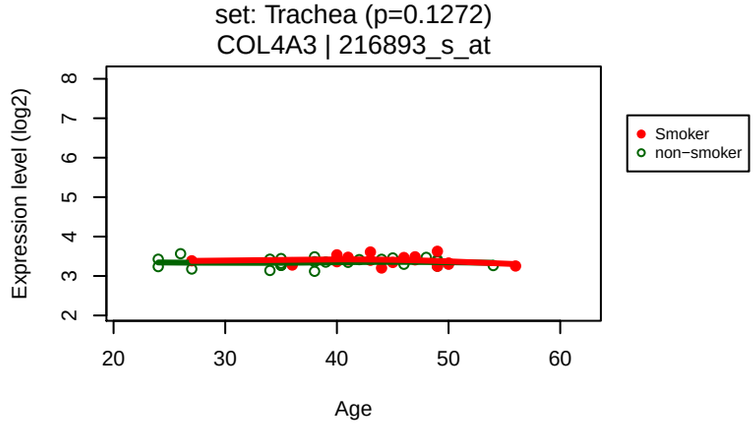

Gene expression ratio in Males
gene: COL4A3 | 216893_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7071
p=0.8224
p=0.2907
p=0.3914
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A3 | 216893_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.976
p=0.6825
p=0.229
p=0.3066
SmallAirway2
LargeAirway2
Trachea
Macrophage2

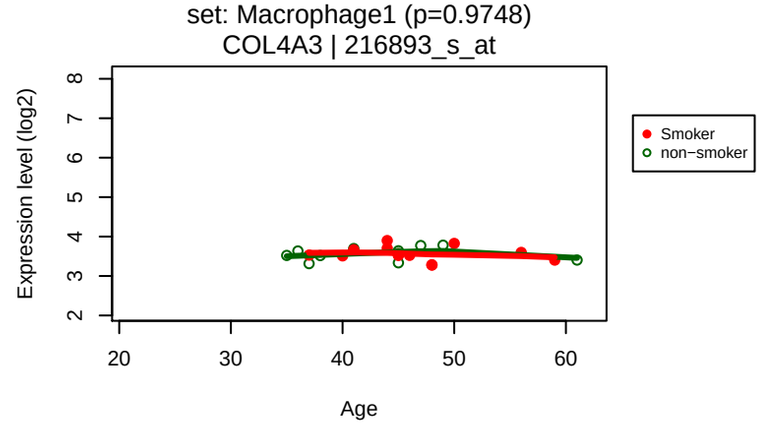

set: SmallAirway1 (p=0.2435)
COL4A3 | 216893_s_at
Expression level (log2)
Age
Smoker
non−smoker

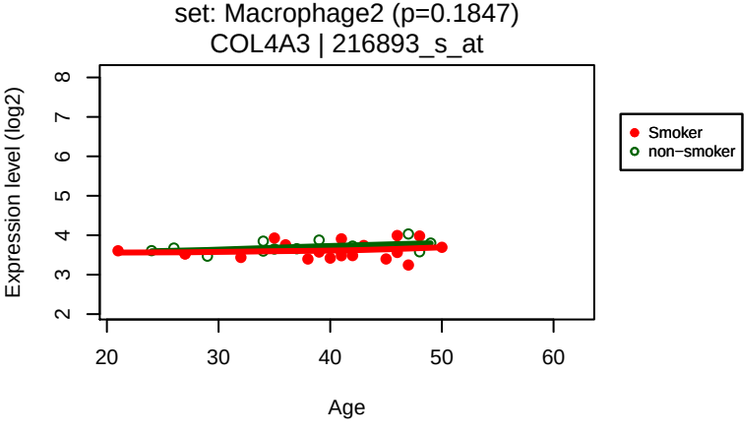

set: SmallAirway2 (p=0.7629)
COL4A3 | 216893_s_at
Expression level (log2)
Age
Smoker
non−smoker

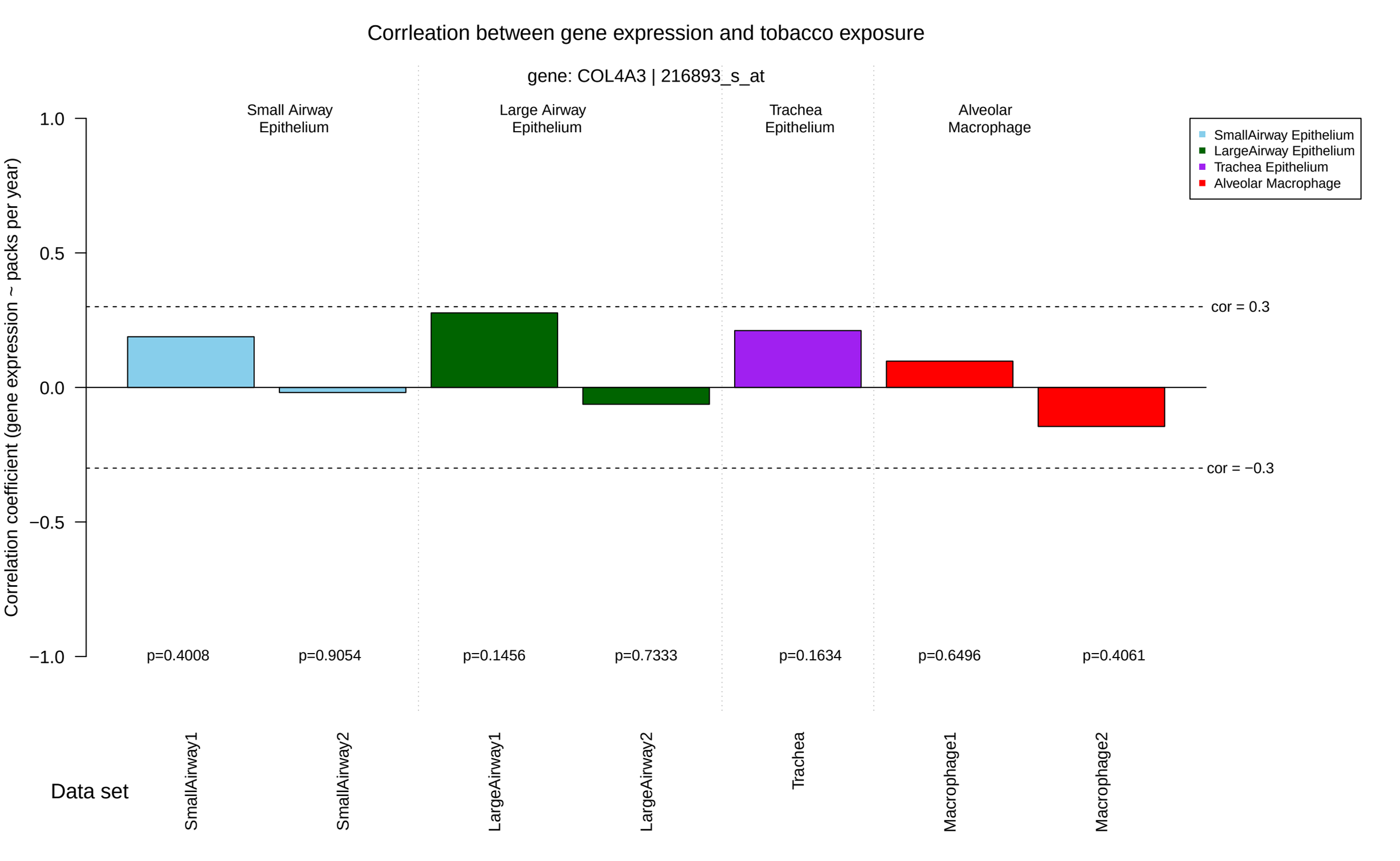

set: LargeAirway1 (p=0.0543)
COL4A3 | 216893_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.652)
COL4A3 | 216893_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1272)
COL4A3 | 216893_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9748)
COL4A3 | 216893_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1847)
COL4A3 | 216893_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A3 | 216893_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4008
p=0.9054
p=0.1456
p=0.7333
p=0.1634
p=0.6496
p=0.4061
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2